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Edp casi lineal o1 sistema caracteristico

Definición 1 (Sistema característico). Sea la EDP casi-lineal de primer orden

$$\forall (x_1, x_2) \in \Omega : a(x_1, x_2, u(x_1, x_2))u_{x_1} + b(x_1, x_2, u(x_1, x_2))u_{x_2} = c(x_1, x_2, u(x_1, x_2)) \qquad (1)$$

donde $a, b, c \colon \mathbb{R}^3 \longrightarrow \mathbb{R}$ son funciones $\mathcal{C}^1$, su sistema característico asociado es

$$\begin{cases} \frac{\partial x}{\partial t} = a(x, y, z) \\ \frac{\partial y}{\partial t} = b(x, y, z) \\ \frac{\partial z}{\partial t} = c(x, y, z). \end{cases}$$

Este es un sistema de ecuaciones diferenciales ordinarias (EDOs) de primer orden donde $x, y, z \colon \mathbb{R} \longrightarrow \mathbb{R}$ son funciones de t.

Referenciado en

- `edp-casi-lineal-o1-curvas-caracteristicas`

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